CMAP Planning Network

Chicago Open DTA dataset · CMAP trip-based model, c25q4 scenario 100 (2019), PM peak 4–6 pm · October 27, 2026; assignment test added November 6, 2026

1 Model fingerprint

Source		**Demand**	
Model	CMAP trip-based, ON TO 2050	Time period	**PM** (period 7, 4–6 pm)
Scenario	**c25q4, 100 = 2019**	Assignment classes	**8**
Emme links (directed)	**55,005**	PM vehicle trips	**3,527,730**
Software	Emme	written (>0.01)	3,481,077 (98.7%)
Driving network		**Coding**	
Directed GMNS links	**55,005** (none dropped)	VDF codes	8
Nodes	**20,190**	Capacity basis	veh/h/lane, LOS E
centroids / road	3,649 / 16,541	VDF	CMAP fd1–fd8
Centroid connectors	7,264	Toll	$ per link, 198 links
Zones	**3,649**	**Output**	
internal / external	3,632 / 17	Columns per link	30
numbering	1–3649, unchanged	CMAP results kept	`ref_volume`, `ref_time`
connected	**all, one strong component**	Gates	**29 pass, 0 fail**

The conversion follows CMAP's own code, not only its documentation. Seven files from CMAP's public model repository settle what the appendix leaves open: `Ftime_capacity.py`, `TOD_network_prep.py`, `vdf_tod.in`, `ttables.mac`, `tod_factors.p7`, `tod_flags.in` and `SOLA_assignment.py`. Every rule, unit and source is listed in `CMAP_network_conversion_reference.pdf`.

2 Reconstruction fidelity

Each test compares a converted value against a value CMAP itself stored or computed. No assignment is involved, so this tests the conversion itself.

Test	**Ours**	**CMAP's stored value**	**Agreement**
`DATA1` vs `@vadt`	55,005 links	max difference 0.0005 veh	**exact**
`TIMAU` vs `timau`	55,005 links	max difference 10^{-6} min	**exact**
Lanes vs `network_p7.txt`	55,005 / 55,005	same node and link counts	**exact**
`@ftime` from binned `@speed`	100% of VDF 1–6, 8	`Ftime_capacity.py` rule	**exact**
VDF 5, 8 at `DATA1` vs `TIMAU`	median −0.04%	`vdf_tod.in` fd5, fd8	**exact**
PM SOV, mf417 + mf427 + mf437	2,351,437	`trip_roster` PM2 mode 1 = 2,351,437	**exact**
OD trips between zones	3,035,518	leaving centroids = 3,065,674	**0.990**

The last test closes the loop between demand and network. The 1% gap is the park-and-ride first legs, which CMAP builds inside its run and does not export. It also settles one fact the guide leaves implicit: **CMAP's stored link volumes are vehicles, not vehicle equivalents**. On a vehicle-equivalent basis the ratio would be 0.957.

The code also settled four points the appendix states differently or not at all:

- `@ftime` rounds `@speed` up to a speed bin (freeway 70 → 65 mph; expressway at least 40);
- the 0.75 capacity factor applies to VDF 1–5, not only arterials;
- `@toll` is dollars per link;
- the emmemat period tables mf94–mf97 and mf14–mf18 hold period 8, not PM.

3 Network and demand, PM

3.1 CMAP's own PM results, by facility

Read from the converted `link.csv` (`ref_volume`, `ref_time`, `vdf_fftt`). Delay is congested minus free-flow vehicle-hours.

VDF	Facility	Links	Lane-mi	VMT veh-mi	VHT veh-h	Delay veh-h	Speed mph
1	Arterial street	41,861	36,969	16,836,841	685,657	234,550	24.6
2	Freeway	1,998	5,512	10,565,352	239,959	59,891	44.0
3	Freeway → arterial ramp	1,992	586	364,490	15,912	3,800	22.9
4	Expressway	930	1,297	1,436,129	40,963	9,399	35.1
5	Freeway → freeway ramp	664	304	338,540	13,349	4,165	25.4
6	Zone centroid connector	7,264	5,052	3,893,909	194,696	0	20.0
7	Toll collection link	188	82	67,954	1,254	0	54.2
8	Metered entrance ramp	108	23	19,478	1,018	241	19.1
	PM, 4–6 pm	**55,005**	**49,824**	**33,522,694**	**1,192,807**	**312,046**	**28.1**

Without connectors, the road network carries 29.6 M VMT at 29.7 mph. In CMAP's PM run, 3,615 road links are above 1.0 of their VDF capacity, and 2,947 run at more than three times their free-flow time (Figure 1).

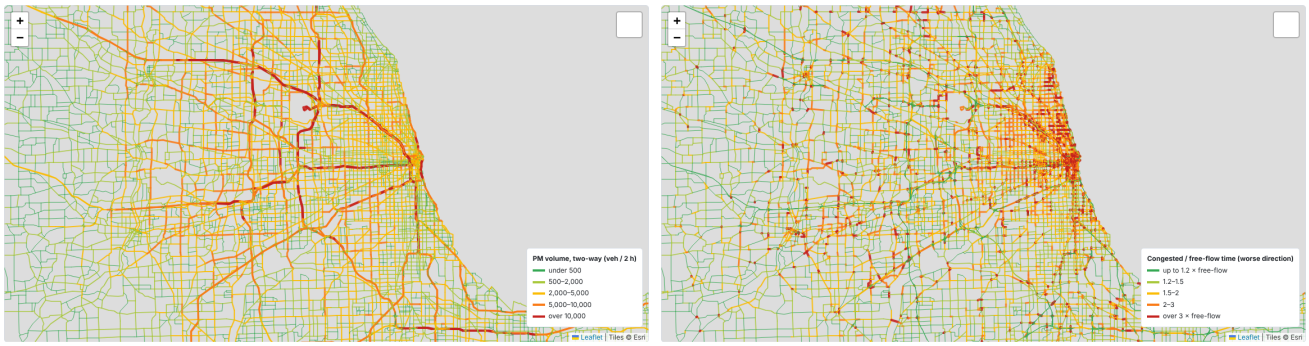


Figure 1: CMAP's PM assignment, central part of the region, connectors not drawn. Left: two-way volume. Right: congested over free-flow time, worse direction. Interactive version with the Esri basemap: `CMAP_report.html`.

3.2 Demand

The PM tables are built the way CMAP's `ttables.mac` builds its period-7 tables. The factors come from `tod_factors.p7`, which equal the data guide's TOD7 column. There is one file per class, in vehicles per 2-hour period.

Class	Source	PM vehicles	OD rows	Kept
`sov`	mf417 + mf427 + mf437 + 0.155×mf8 (external autos)	2,393,513	818,885	100.0%
`hov2`	mf447	513,112	342,932	100.0%
`hov3`	mf457	141,612	204,014	100.0%
`apv`	0.155×mf10, airport zones only	11,839	26,944	99.9%
`btrk`	0.176×mf4 (B-plate)	307,563	1,338,781	96.5%
`ltrk`	0.176×mf5	51,627	718,863	71.9%
`mtrk`	0.176×mf6; pce 2	48,066	664,820	70.9%
`htrk`	0.103×(mf7 + mf9 external trucks); pce 3	60,398	659,707	87.6%
	total	**3,527,730**	**4,774,946**	**98.7%**

Demand floor. The truck tables are dense: about 13 million positive cells each, most of them a small fraction of a trip. Cells at or below **0.01 vehicles** are not written and are not rescaled. Autos keep over 99.9%. Light and medium trucks keep about 71%, and heavy trucks 88%.

4 Assignment run

First run, preliminary (November 6, 2026). TAPLite assigned the converted PM network and demand with one BPR function on every road link. Inputs, the TAPLite build, run scripts and the evaluation are in `city-tensor/cmap_ta_bpr_allmodes`.

Setting	Value
Network, demand	this conversion: 55,005 links, 3,649 zones, 3,481,077 PM vehicles (all written cells)
Classes	5: `car` = sov + hov2 + hov3 + apv; `btrk`, `ltrk`; `mtrk` (pce 2); `htrk` (pce 3)
Volume-delay	BPR, $\alpha = 0.15$, $\beta = 4$, on `@emcap` × lanes; connectors and toll links at `@ftime`
Tolls, bus preload	not included
Run	10 iterations, relative gap 0.18 %, 7 min 43 s on 2 cores, 5.3 GB memory

The four car modes are merged into one class because eight classes did not fit in 8 GB (about 8.5 GB needed). Without tolls the four car modes see the same costs, so link volumes are unchanged; only the split by car mode is lost.

Facility	**VMT (M)**		**VHT (k)**		**Speed (mph)**	**Volume**
	TAPLite / CMAP	ratio	TAPLite / CMAP	ratio	TAPLite / CMAP	R^2
Arterial street	15.38 / 16.84	0.91	427.8 / 685.7	0.62	36.0 / 24.6	0.86
Freeway	9.96 / 10.57	0.94	181.0 / 240.0	0.75	55.0 / 44.0	0.92
Expressway	1.12 / 1.44	0.78	24.8 / 41.0	0.61	45.2 / 35.1	0.72
Fwy → art. ramp	0.36 / 0.36	0.99	12.7 / 15.9	0.80	28.4 / 22.9	0.62
Fwy → fwy ramp	0.30 / 0.34	0.88	8.7 / 13.3	0.65	34.3 / 25.4	0.74
Toll collection	0.07 / 0.07	1.04	1.3 / 1.3	1.05	53.8 / 54.2	0.90
Metered ramp	0.02 / 0.02	1.14	1.0 / 1.0	1.02	21.3 / 19.1	0.61
Road links	**27.22 / 29.63**	**0.92**	**657.4 / 998.1**	**0.66**	**41.4 / 29.7**	**0.93**

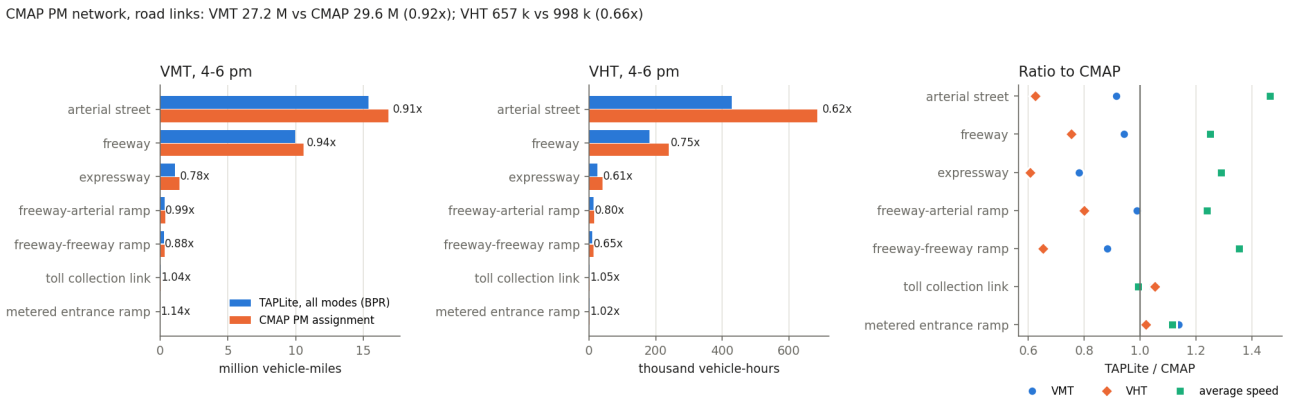


Figure 2: TAPLite (BPR 0.15, 4) against CMAP's PM assignment by facility type, road links.

Reading. Route choice is right: volumes follow CMAP on freeways and arterials (VMT 0.94 and 0.91, R^2 0.92 and 0.86). Delay is not: TAPLite's network delay is 34 thousand vehicle-hours against CMAP's 312 thousand, so speeds are 25–46 % higher. Where CMAP has no delay the link times agree; where CMAP is congested, BPR (0.15, 4) on `@emcap` adds much less. CMAP's functions are steeper (0.75 × capacity, $\beta = 8$ on freeways) and add signal delay on arterials (arterial median time ratio 0.77). Expressways carry 0.78 of CMAP's traffic because the run has no tolls, so the parallel tollways attract more.

Two choices are still open:

- **VDF form.** The BPR test above is too flat. The next candidates are BPR with CMAP's own α, β and the 0.75 capacity factor, or a BPR fitted to each CMAP curve (fd1–fd8). CMAP's min-term (fd2, fd4) and Webster signal delay (fd1, fd3) are not TAPLite functions.
- **Bus preload.** CMAP counts buses as 3 vehicle equivalents each (`el2`). These are not exported, so the runs so far are without them.

5 Model comparison, PM period

All links including connectors, 4–6 pm. "Ours" is the first TAPLite run of Section 4 (BPR 0.15, 4; no tolls). Delay is congested minus free-flow vehicle-hours.

Measure	Ours	CMAP	Ratio	Reading
Demand (vehicle trips between zones)	3,035,518	3,065,674	**0.990**	park-and-ride legs missing
Link volume (vehicles)	54,301,914	56,908,112	**0.954**	
VMT (vehicle-miles)	31,016,448	33,522,694	**0.925**	route choice close
VHT (vehicle-hours)	847,447	1,192,807	0.710	BPR too flat
Delay (vehicle-hours)	33,952	312,046	**0.109**	BPR too flat
Average speed (mph)	36.6	28.1	1.302	
Link volume R^2	road links		**0.93**	

6 Open items

	Issue	What would resolve it
O1	CMAP's arterial function adds Webster delay from the cycle length and g/C at the downstream node (`@cycle`, `@gc`, packed into `ul3`). Neither is in the export, so arterial times sit 22% below `TIMAU` at the median.	The values from CMAP's emmebank, or a rerun of `Arterial_delay.py` on the exported network.
O2	CMAP's link volume includes `el2` (buses at 3 veq each), which the PM files do not carry. Freeway and expressway times reproduce at −9.5% and −1.3% without it.	`busveq` from the emmebank, or rebuilt from the transit itineraries.
O3	Park-and-ride first legs (mf59) are built inside CMAP's run. They are the 1% gap in the demand loop.	The convolution in `ttables.mac`, run on mf837 and the transit tables.
O4	The 0.01 floor drops 28–29% of light and medium trucks.	A per-origin rescale of the kept cells, or a lower floor (99.7% kept at 0.0001).
O5	Scenario 100 is 2019; the speed data is 2023.	`c25q4_200` (2025) has the same file structure, and the same code converts it.
O6	BPR (0.15, 4) gives 0.11 of CMAP's delay and 0.71 of its VHT.	CMAP's own α, β with the 0.75 factor, or a BPR fitted to each of fd1–fd8.
O7	No tolls in the first run; expressways carry 0.78 of CMAP's VMT. Eight classes need about 8.5 GB.	Toll as link cost with each class's value of time; run all eight classes on a 16 GB machine.

7 Observation alignment and next steps

Alignment with real-world observations is partly possible already, which is where Chicago differs from SCAG. The dataset holds 2023 INRIX TMC speeds, hourly, on 18 corridor-directions, with fitted QVDF parameters per TMC. These are mapped to the OSM City-of-Chicago network today; the next step maps the same TMCs to

CMAP links. **Counts are still missing**: every corridor demand is derived from speed, and IDOT AADT is the first independent volume check to add.

	Stage	What it provides	Status
1	Standardized, documented network	every `link.csv` column with source, unit and reference; every CMAP number in one region file	**complete (PM)**
2	Open executable assignment	TAPLite replay of CMAP's PM assignment	first run (BPR)
3	Observation alignment	TMC speeds and QVDF on CMAP links; AADT as the volume check	speeds in hand
4	AI applications	multi-day DTA benchmark (OpenDTA), flow-through tensor, bottleneck detection, scenario screening	planned

	Next step
1	Rerun the TAPLite PM replay with CMAP-shaped VDFs and tolls (O6, O7); compare volume and time link by link
2	Map the 18 corridor-directions of TMCs onto CMAP links
3	Add IDOT AADT as the first independent volume check
4	Resolve O1–O4 where the emmebank allows
5	Convert AM (period 3) and the 2025 scenario with the same code

Sources: CMAP Travel Demand Model Data User Guide (c25q4), CMAP Data Hub; ON TO 2050 travel demand model appendix; CMAP trip-based model code (commit e52b318).